

Aspirin: Mechanism of Action

Aspirin's ability to suppress the production of prostaglandins and thromboxanes is due to its irreversible inactivation of the cyclooxygenase enzyme required for prostaglandin and thromboxane synthesis. Aspirin acts as an acetylating agent where an acetyl group is covalently attached to a serine residue in the active site of the COX enzyme. This makes aspirin different from other NSAIDs (such as diclofenac and ibuprofen), which are reversible inhibitors.

Low-dose aspirin use irreversibly blocks the formation of thromboxane A₂ in platelets, which inhibits platelet aggregation during the lifetime of the affected platelet (8-9 days). This antithrombotic property makes aspirin useful for reducing the incidence of heart attacks in people who have had a heart attack, unstable angina, ischemic stroke or transient ischemic attack. 40 mg of aspirin a day is able to inhibit a large proportion of maximum thromboxane A₂ release provoked acutely, with the prostaglandin I₂ synthesis being little affected; however, higher doses of aspirin are required to attain further inhibition.

Prostaglandins, a type of hormone, have diverse effects, including the transmission of pain information to the brain, modulation of the hypothalamic thermostat, and inflammation. Thromboxanes are responsible for the aggregation of platelets that form blood clots. Heart attacks are caused primarily by blood clots, and low doses of aspirin are seen as an effective medical intervention to prevent a second acute myocardial infarction.

At least two different types of cyclooxygenases, COX-1 and COX-2, are acted on by aspirin. Aspirin irreversibly inhibits COX-1 and modifies the enzymatic activity of COX-2. COX-2 normally produces prostanoids, most of which are proinflammatory. Aspirin-modified COX-2 produces epi-lipoxins, most of which are anti-inflammatory. Newer NSAID drugs, COX-2 inhibitors, have been developed to inhibit only COX-2, with the intent to reduce the incidence of gastrointestinal side effects.

Several COX-2 inhibitors, such as rofecoxib, have been withdrawn from the market, after evidence emerged that COX-2 inhibitors increase the risk of heart attack and stroke. Endothelial cells lining the microvasculature in the body are proposed to express COX-2, and, by selectively inhibiting COX-2, prostaglandin production is downregulated with respect to thromboxane levels, as COX-1 in platelets is unaffected. Thus, the protective anticoagulative effect of PGI₂ is removed, increasing the risk of thrombus and associated heart attacks and other circulatory problems.

Aspirin has been shown to have at least three additional modes of action. It uncouples oxidative phosphorylation in cartilaginous and hepatic mitochondria, by diffusing from the inner membrane space as a proton carrier back into the mitochondrial matrix, where it ionizes once again to release protons. Aspirin buffers and transports the protons. When high doses are given, it may actually cause fever, owing to the heat released from the electron transport chain, as opposed to the antipyretic action of aspirin seen with lower doses.